

SRI LANKA INSTITUTE OF
ADVANCED TECHNOLOGICAL
EDUCATION



Higher National Diploma in Information Technology
Advanced Technological Institute Dehiwala

Hostel Room Allocation & Complaint System

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Statement of Originality

I hereby declare that the assignment entitled “**Hostel Room Allocation & Complaint System**”, submitted to the **Department of Information Technology, Advanced Technological Institute in Dehiwala**, is our original work carried out by us as part of our academic requirements.

This assignment has not been submitted previously, either in part or in whole, to any other institution or examination body for any degree or diploma. All the information presented in this document has been prepared in accordance with academic integrity and ethical standards. Any external sources used for reference purposes have been appropriately acknowledge.

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Abstract

The Hostel Room Allocation & Complaint System is designed to improve the management of hostel facilities in educational institutions. Traditional hostel management methods rely heavily on manual records, which often result in inefficiencies, data inaccuracies and delays in handling student issues.

This system focuses on managing hostel related data such as student details, hostel and room information, room allocation records and student complaints through a centralize database. The proposed database system ensures accurate data storage, easy retrieval and effective management of hostel operations.

By implementing this system, hostel administrators can efficiently allocate rooms, track occupancy and handle student complaints in a structured and reliable manner. This proposal outlines the objectives, database structure and expected benefits of the hostel room allocation & complaint system.

Background

Hostel play a vital role in providing accommodation facilities to students in educational institutions. Managing a large number of students, rooms and complaints manually can be time consuming and prone to errors. Paper based systems make it difficult to track room availability, allocation history and unresolved complaints.

As the number of students increases, maintaining accurate records becomes more challenging. Delays in room allocation and complaint resolution can negatively affect student satisfaction and hostel administration efficiency. These challenges highlight the need for a computerized database system that can organize and manage hostel related data effectively.

Problem Statement

The existing manual hostel management system faces several issues:

- Difficulty in tracking room availability and allocation.
- Inaccurate or incomplete student records.
- Delays in handling student complaints.
- Data redundancy and inconsistency.
- Time consuming manual record maintenance.

Objectives and Aims

❖ Objectives

- ✓ To design a centralized database for hostel management.
- ✓ To maintain accurate student and hostel records.
- ✓ To manage room allocation and track student complaints.
- ✓ To reduce manual paperwork and errors.

❖ Aims

- ✓ To improve efficiency in hostel administration.
- ✓ To ensure data accuracy and consistency.
- ✓ To support quick retrieval of hostel related information.
- ✓ To provide a scalable database design for future expansion.

Database Design Overview

❖ Include tables

- Student
- Hostel
- Room
- Allocation
- Complaint
- Warden

❖ Structure overview

- A hostel contains multiple rooms.
- Each student is allocated to a room.
- A room can be allocated to different students over time.
- A student can submit multiple complaints.
- Each hostel is supervised by a warden.

❖ Cardinalities

- Hostel – Room : 1 to Many
- Room – Allocation : 1 to Many
- Student – Allocation : 1 to 1
- Student – Complaint : 1 to Many
- Warden – Hostel : 1 to 1

Resource Utilization

- ❖ Database tool → MySQL
- ❖ Diagram tool → draw.io
- ❖ Learning resources → lecture notes, online tutorials and Navathe's book

Methodology

1. Identification of entities and relationships.
- 2.Design of ER diagram with cardinalities and participation constraints.
- 3.Conversion of ER diagram into relational schema.
- 4.Implementation of SQL queries for table creation and data insertion.
- 5.Validation of database design.

Conclusion

The hostel room & complaint system provides an effective solution for managing hostel operations using a structured database design. By digitalizing hostel records, the system reduces manual effort, improves data accuracy and enhances administrative efficiency.

The project demonstrates the practical application of database concepts such as entities, relationships, cardinalities and constraints. The proposed system is suitable for academic purposes and can be extended to real world hostel management applications in the future.